

ESP32-S3 Water Potability & Treatment Guidelines

Technical Knowledge Base & RAG Grounding Manual | Version 2.4

Operational Purpose: Comprehensive specification document establishing parameter thresholds, deterministic potability evaluation logic, root-cause diagnostics, hardware telemetry protocols, and automated treatment pipeline architectures optimized for ESP32-S3 TFT displays.

1. Primary Parameter Thresholds & Operational Ranges

Water telemetry is continuously evaluated against three primary physical and chemical parameters: Potential of Hydrogen (pH), Total Dissolved Solids (TDS), and Electrical Conductivity (EC).

1.1 pH Classification Spectrum (1.0 to 14.0)

- **Potable (Ideal):** 6.5 to 8.5 (Balanced neutral state)
- **Acidic Range:** 1.0 to 6.49 (Corrosive risk)
- **Alkaline Range:** 8.51 to 14.0 (Scaling/caustic risk)

1.2 TDS Classification (mg/L or ppm)

- **Low / Flat:** 0.0–49.9 (Mineral deficient)
- **Ideal Quality:** 50.0–300.0 (Optimal balance)
- **Good Quality:** 300.1–500.0 (Acceptable tap)
- **Poor Quality:** 500.1–1000.0 (Mineral heavy)
- **Unacceptable:** > 1000.0 (Brackish/Saline)

1.3 Electrical Conductivity - EC (µS/cm)

Core conversion mathematical formula: $EC = TDS / 0.65$

- **Ideal Range:** 0.0 to 400.0 µS/cm (Low mineral conductivity)
- **Acceptable Range:** 400.1 to 800.0 µS/cm (Standard municipal water)
- **High Range:** 800.1 to 1500.0 µS/cm (Elevated ionic concentration)
- **Severe Range:** > 1500.0 µS/cm (High salinity contamination)

2. Potability Evaluation Logic & Decision Matrix

Deterministic Rule: To classify as **POTABLE**, water telemetry must satisfy **ALL conditions simultaneously**: pH between 6.5 and 8.5, TDS between 50.0 and 500.0 mg/L, and EC between 0.0 and 800.0 µS/cm. Any single parameter violation immediately flags the water as **NOT POTABLE**.

Root Cause Analysis Matrix: pH Deviations

Granular diagnostic evaluation for acidic and alkaline water anomalies

3.1 Granular pH Evaluation Matrix

Violation Category	Range	Cause (Sentence Format)	Recommendation (Sentence Format)
Extremely Acidic	1.0–1.99	The water is extremely acidic, causing severe metal leaching.	Install a sodium hydroxide dosing system to neutralize the acid.
Highly Acidic	2.0–2.99	The water is highly corrosive and dissolves copper and lead.	Install a calcite and soda ash neutralizing tank.
Strongly Acidic	3.0–3.99	The water has high acidity which poses a severe pipe damage risk.	Install a soda ash injection pump followed by an RO unit.
Acidic	4.0–4.99	The water has a sour taste and causes gradual pipe wear.	Pass the water through a dual-tank limestone bed.
Mildly Acidic	5.0–5.99	The water is mildly acidic, causing a metallic taste and slow corrosion.	Install a calcite media filter to raise the pH.
Slightly Acidic	6.0–6.49	The water is slightly below the safe limit and mildly corrosive.	Install an inline remineralization cartridge.
Slightly Alkaline	8.51–8.99	The water has high bicarbonate levels and reduces chlorine effectiveness.	Install a cation exchange resin filter.
Alkaline	9.0–9.99	The water has a bitter taste and causes mineral scale buildup.	Install a weak acid softener system.

Root Cause Analysis Matrix: Alkalinity & Mineralization

Advanced diagnostic tables for extreme pH and high TDS conditions

3.1 Granular pH Matrix (Continued)

Violation Category	Range	Cause (Sentence Format)	Recommendation (Sentence Format)
Strongly Alkaline	10.0–10.99	The water causes heavy scaling and presents a high pH hazard.	Install an HCl acid dosing pump to lower the pH.
Highly Alkaline	11.0–11.99	The water is a chemical hazard with tissue damage risks.	Install an acid dosing system and an RO unit.
Extremely Alkaline	12.0–12.99	The water is caustic and toxic to consume.	Implement a heavy hydrochloric acid dosing loop.
Dangerously Caustic	13.0–13.99	The water is at bleach level and causes skin irritation.	Shut the main valve immediately and dose with acid.
Maximum Alkalinity	14.0	The water is at lye level and is highly corrosive to all tissue.	Trigger system lockout and treat as hazardous waste.

3.2 TDS & EC Evaluation Rules

Violation Category	TDS / EC Range	Cause (Sentence Format)	Recommendation (Sentence Format)
Very Low Minerals	TDS < 50 & EC < 75	The water lacks essential mineral ions, resulting in a flat taste.	Install a calcium and magnesium remineralization cartridge.
High Minerals	TDS 500–1000 & EC 800–1500	The water is hard and causes noticeable scale buildup.	Install a 100GPD reverse osmosis membrane system.
Severe TDS/EC	TDS > 1000 & EC > 1500	The water is brackish and unsafe for human consumption.	Install an industrial water distiller or multi-pass RO system.

Multi-Parameter Logic, Diagnostics & Treatment Pipelines

System conflict resolution, ATC formulas, fault codes, and treatment architectures

4. Multi-Parameter Conflict Resolution

- **High TDS + Low pH** (e.g., pH 5.4, TDS 850): *Cause:* The water contains high dissolved solids that are acidic in nature. *Action:* Install a reverse osmosis unit followed by an alkaline post-filter.
- **High TDS + High pH** (e.g., pH 9.2, TDS 1100): *Cause:* The water has high TDS composed of alkaline scaling minerals. *Action:* Install a cation softener followed by a reverse osmosis unit.
- **Normal TDS + Extreme pH** (e.g., pH 3.5, TDS 120): *Cause:* The water suffers from severe chemical acid contamination. *Action:* Install a sodium carbonate dosing pump to normalize the pH.

5. Granular TDS Quality & Taste Stratification

- **0.0–49.9 ppm:** Flat taste. *Action:* Add remineralizer. | **50–150 ppm:** Ideal. *Action:* None.
- **150.1–300 ppm:** Tap quality. *Action:* None. | **300.1–500 ppm:** Hard limit. *Action:* Carbon block.
- **500.1–1000 ppm:** Salty/scaling. *Action:* RO system. | **1000.1–2000 ppm:** Brackish. *Action:* Dual-pass RO.
- **> 2000 ppm:** Saline (toxic). *Action:* Thermal distiller.

6. Temperature Compensation & 7. Sensor Diagnostics

Baseline: 25 °C | $EC_{25} = EC_{\text{measured}} / (1 + 0.02 \times (T_{\text{measured}} - 25))$ | $TDS = EC_{25} \times 0.65$

- **Fault 1 (Dry Probe):** pH = 0.0–0.5 AND EC = 0.0 → "SENSOR FAULT: Probes are dry."
- **Fault 2 (Logic Conflict):** TDS > 0.0 BUT EC = 0.0 → "CALIBRATION FAULT: Check EC wiring."
- **Fault 3 (Out of Bounds):** pH < 0.0 OR pH > 14.0 → "CALIBRATION FAULT: Recalibrate pH."

8. Standardized Treatment Pipeline Architectures

- **Pipeline A (pH < 6.5):** Sediment Filter → Calcite Tank → Carbon Block → UV Lamp
- **Pipeline B (TDS > 500):** Sediment Filter → Carbon Block → RO Membrane → Remineralizer
- **Pipeline C (pH > 8.5):** Sediment Filter → Cation Softener → RO Membrane → Remineralizer